PROBLEM STATEMENT

Global Insure, a leading insurance company, processes thousands of claims annually. However, a significant percentage of these claims turn out to be fraudulent, resulting in considerable financial losses. The company's current process for identifying fraudulent claims involves manual inspections, which is time-consuming and inefficient. Fraudulent claims are often detected too late in the process, after the company has already paid out significant amounts. Global Insure wants to improve its fraud detection process using data-driven insights to classify claims as fraudulent or legitimate early in the approval process. This would minimise financial losses and optimise the overall claims handling process.

METHODOLOGY

- The model used to provide resolution is Logistic Regression with RFECV to choose the best variables and Random Forests with GridSearchCV to hyper tune the parameters.
- We have used the above 2 algorithms as this is a classification task.
- For Logistic regression, we have used the RFECV to choose the best 'n' features and then built a model using the logistic regression from statsmodel to get information on the p values to understand the significance of all the selected features and to understand their relationship with the target variable.
  We have also used variance inflation factor to identify if there's any multicollinearity among the selected features.
- For Random Forest, we built a model by selecting few features based on the feature importances returned by Random Forest and then further enhanced the model by hyper tuning the parameters using GridSearch CV.
- To evaluate the performance of the model, we have used the sensitivity, specificity, recall, precision, F-1 score and accuracy

TECHNIQUES

- **<u>Missing Data Treatment:</u>**
  - We treated the missing data from the dataset before moving forward.
  - The data had 2 columns with null values - authorities contacted and _c39 column.
  - "Authorities contacted" was treated for NULL values whereas _c39 was a completely empty column, which is why it was dropped.
- **<u>Redundant Values/Values needing treatment:</u>**
  - "?" was present in these columns which needed to be treated - collision type, property damage, police_report_available.
  - For collision type, this character was replaced by a valid value based on its dependency on another variable but for the remaining 2 columns - property damage, police_report_available, we replaced the value with "NO", assuming

that there was no property damage reported, or no police report was available for these records.
    - We also mapped few redundant values which could be grouped together in insured hobbies and incident severity columns.

- **Invalid Value Treatment:**
    - We did a sanity check on all the numerical columns and found that the umbrella_limit column had a negative value which needs to be dropped.

- **High Cardinality Columns:**
    - We dropped the columns wherein the cardinality was greater than 90%, which means that for majority of the records, the value is unique per row and such variables have very limited predictive power.
    - We dropped 4 such variables - policy_number, policy_bind_date, policy_annual_premium, insured_zip

- **Fix Data Types:**
    - There was one column 'incident_date' column which needed to be converted to datetime from object

- **Train-Test Split:**
    - The data was split into training and test sets in 70- 30 proportion and stratification was used on the target variable as there was a class imbalance.

- **Exploratory Data Analysis:**
    - **Univariate Analysis of Numerical columns :**
      We looked into the distributions and boxplots of all the numerical columns to identify skewness and outliers and found that there were outliers in the total_claim_amount,property_claim, injury_claim and vehicle claim variables, which should not be treated as they can be an indicator for frauds.
      On analysing the histograms , we observed that the “umbrella_limit” variable has majority of values as zero, which is not a very meaningful insight, which is why we dropped this variable later.

    - **Bivariate Analysis of Numerical columns :**
      We performed the bivariate analysis of the numerical columns by creating the correlation matrix and found that Age has a very high +ve correlation with months_as_customer and total_claim_amount has a high +ve correlation with property_claim, injury_claim and vehicle claim.

    - **Target Likelihood Analysis:**
        - **Numerical Columns :**

We plotted the box plots for each of the numerical columns for both “Y” and “N” outcome to understand the data spread.

- **Categorical Columns:**
  We calculated the likelihood for each level of every relevant categorical feature for the target = “Y” event and also plotted the bar plots accordingly.

- **Class Imbalance Treatment:**
  As there was a class imbalance in the dependent variable, we used RandomOverSampler to resample the training and testing data.

- **Feature Engineering:**
  - **New Feature Derivations:**
    Based on the EDA done in the previous steps, we derived new features by
    - segmenting customers based on the months_as_customer,
    - calculated net gain from capital loss and capital gain
    - calculated the claim% for vehicle, property and injury claims
    - grouped education level, relationship and hobbies into categories
    - derived incident year and incident month from the incident_date

  - **Redundant columns:**
    After deriving new features, we dropped these features as:
    - Highly correlated –
      Injury_claim, property_claim, vehicle_claim,age
    - Irrelevant/Redundant -
      umbrella_limit, collision_type, incident_hour_of_the_day
    - Info already extracted -*insured_education_level, insured_relationship,insured_hobbies, months_as_customer,incident_date,capital-gains,capital-loss,total_claim_amount*

  - **Grouping values in Categorical columns:**
    We grouped values from auto_model, insured occupation, incident type and incident state columns to make the categories more significant from a predicting standpoint.

  - **Dummy variable creation:**
    Dummy variables were created for all the categorical variables and as the target variable was also of object datatype, we mapped them to 1’s and 0’s.

  - **Feature Scaling:**

Once all the variables were converted to numeric, we scaled all the features using Standard Scaler.

- **Model Building & Evaluation:**
  - **Logistic Regression:**
    - Logistic Regression model was built using scikit-learn and was passed to RFECV to get the best features. The RFECV returned 7 features with Ranking = 1 which was used to further build a logistic regression model using statsmodel
    - On building the logistic regression model using statsmodels with the RFECV returned features and checked the p-values for all these features and found that the p-value is less than 0.05 for all the variables, stating the significance of all the selected variables.
    - The VIF was also less than 5 for all the variables.
    - Predictions were made using the statsmodel model with cutoff = 0.5 and the accuracy came out to be 83.37% on training set.
    - To get better metric values for Specificity, Sensitivity, Recall and Precision, we looped through different cutoff values to get the optimal cutoff.
    - ***Based on the Accuracy, Sensitivity and Specificity trade off curve and the Precision-Recall curve, the optimal cutoff came out as 0.6 and after making predictions on this cutoff, the final accuracy on training dataset = 81.74% and on the testing dataset = 80.66%.***

  - **Random Forest:**
    - Random Forest model was built from scikit learn based on feature importances returned by the Random Forest model.
    - The training accuracy came out as 99.8% on the training dataset and 71% on the test dataset, which showed that the model was overfitting.
    - To improve the model performance, we used GridSearchCV to hyper tune the parameters.

    - **On making the predictions on the training dataset with the tuned parameters, the accuracy come out as = 82.8% and on the test data, the accuracy came out as 80.66%, which solved the issue of overfitting.**

INSIGHTS

- Based on the accuracy scores, we can go for either of the models – logistic regression or Random Forest, but I would go with Logistic regression because of the interpretability of the model.
- Based on the coefficient values for the selected variables for the logistic regression model,
  The most significant variable is insured_hobbies_grp_NON-ADVENTURE - BOARD GAMES which basically signifies those hobbies like Chess especially are highly prone to fraud related to claims.
  incident_severity_Minor Damage and incident_severity_Total Loss have a negative correlation coefficient which signifies that they are inversely proportionate to the frauds, stating that people who experience minor damages or total loss and less likely to apply for fraud claims.
- In the Random Forest model, a different set of columns were selected based on the importances.

RECOMMENDATIONS

- Claims from hobbies like chess or board games should undergo additional verification
- Minor damage claims or total loss claims should not go through a rigorous process of review due to low fraud probability.
- Cases with major damage should be investigated more by the fraud detection resources.